Dynamic Risk Intelligence Engine: A Novel Hybrid Optuna-Tuned Ensemble Architecture with Dual-Layer Model-Agnostic Explainability (XAI) for Automated Credit Underwriting

P. Sai Akshitha[A], Sushmita Sallam[B], Katneni Nithya[C], Syed Zuvaria[D]

[A]Department of Artificial Intelligence & Data Science,
Koneru Lakshmaiah Education Foundation, Bowrampet, Hyderabad, Telangana, India

[B]Department of Computer Science and Information Technology,
Koneru Lakshmaiah Education Foundation, Bowrampet, Hyderabad, Telangana, India

***Abstract*—Financial institutions face a persistent challenge in balancing the efficiency of automated credit underwriting with strict regulatory requirements for model transparency. Traditional linear risk scoring models, while interpretable, often fail to capture the complex, non-linear relationships inherent in modern financial data. In contrast, complex machine learning ensemble models achieve high predictive accuracy but lack transparency, making it difficult for human loan officers to explain rejection decisions to auditors and applicants. This paper presents an end-to-end explainable machine learning framework for credit risk assessment and early warning detection, addressing this trade-off. Evaluated on the benchmark UCI Statlog German Credit Dataset (1,000 applicant records) and cross-validated on the Give Me Some Credit Dataset (150,000 records), the proposed pipeline integrates categorical encoding, custom financial ratio engineering, and Synthetic Minority Over-sampling Technique (SMOTE) to mitigate class imbalance without introducing data leakage. Hyperparameters across five baseline and ensemble algorithm families were dynamically optimized using Optuna Bayesian search, resulting in a Random Forest model that achieved an overall accuracy of 81.0% and an ROC-AUC of 81.3%, outperforming traditional statistical models and state-of-the-art neural architectures (TabNet and Deep MLP). To satisfy regulatory transparency requirements, a dual-layer Explainable AI (XAI) engine was incorporated, combining SHAP and LIME with an Automated Risk Action Matrix and a natural language AI assistant.**

I. Introduction

A. Background

Credit risk management is a critical operational component for global lending institutions, relying on the analysis of applicant financial profiles to assess default probability. Traditional statistical techniques, such as Logistic Regression and Discriminant Analysis, have been widely used for credit scoring. However, these methods are limited by their assumptions of linearity and independence, which are often violated in modern financial data characterized by strong non-linear interactions [1], [2]. As a result, traditional linear models frequently produce suboptimal risk classifications and higher rates of misclassified defaults.

II. Related Work

A systematic review of existing literature demonstrates the ongoing tension between predictive performance and model interpretability across financial risk assessment domains.

III. Literature Survey

Recent advancements in Explainable AI (XAI) have focused on model-agnostic techniques that approximate complex non-linear decision boundaries post-hoc. Game-theoretic approaches, such as SHapley Additive exPlanations (SHAP), compute the marginal contribution of each feature across all possible feature subsets [3]., guaranteeing mathematical consistency. Simultaneously, local surrogate models, such as Local Interpretable Model-agnostic Explanations (LIME), construct local linear models around specific data points to generate human-readable decision rules [4]. While SHAP provides population-level feature rankings and LIME provides local instance explanations, existing literature rarely integrates both into an automated risk action matrix for real-time credit underwriting.

IV. Background And Preliminaries

This section outlines the foundational algorithms and mathematical formulations used throughout the proposed framework.

A. XGBoost Regularized Objective Function

The XGBoost regularized objective function is expressed as:

$$\mathcal{L}^{(t)} = \sum_{i=1}^{N} l(y_i, \hat{y}_i^{(t)}) + \sum_{k=1}^{t} \Omega(f_k) \tag{1}$$

where l represents the loss function, y_i denotes the actual target value, $\hat{y}_i^{(t)}$ represents the predicted value at iteration t, and $\Omega(f_k)$ denotes the regularization term used to control model complexity and prevent overfitting [5].

TABLE I
LITERATURE COMPARISON

Study / Reference	Core Methodology Used	Primary Strengths	Identified Weakness / Research Gap
Traditional Scoring [1]	Logistic Regression & Discriminant Analysis	Fully interpretable, simple math	Low accuracy on non-linear data
Theodorakopoulos et al. [1]	Big Data Systematic Risk Review	High coverage across risk domains	Neglects local surrogate XAI rules
Shi et al. [2]	Deep Learning (Bi-LSTM + Attention)	Captures sequential transactions	Opaque black-box, high compute cost
Bussmann et al. [4]	XGBoost + SHAP Explainability	Provides global feature rankings	Lacks local surrogate rule engine
Proposed Framework	Optuna Ensemble + Dual XAI (SHAP/LIME)	81.3% AUC, full compliance	Evaluated on tabular financial data

1) SHAP Additive Feature Attribution Value: SHAP uses game-theoretic Shapley values to calculate the contribution of individual features to model predictions [3].

$$\phi_i(x) = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|!(|F| - |S| - 1)!}{|F|!} [f_x(S \cup \{i\}) - f_x(S)] \tag{2}$$

2) LIME Local Surrogate Loss Minimization: LIME generates local explanations by approximating the predictions of a complex model using an interpretable surrogate model [4].

$$\text{Explanation}(x) = \arg\min_{g \in G} [L(f, g, \pi_x) + \Omega(g)] \tag{3}$$

V. PROBLEM FORMULATION

Let $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^{N}$ represent a credit dataset of N applicants, where $x_i \in \mathbb{R}^d$ represents a d-dimensional feature vector (financial ratios, credit history, demographics) and $y_i \in \{0, 1\}$ represents binary credit risk (0 = Good Risk, 1 = Default Risk). The objective is to learn a predictive mapping function $f : \mathbb{R}^d \to [0, 1]$ that maximizes default discrimination (ROC-AUC) while simultaneously generating an explanation function $\mathcal{E}(f, x_i) \to \mathbb{R}^d$ that provides human-understandable feature attributions compliant with regulatory mandates.

VI. METHODOLOGY

The proposed financial risk intelligence framework follows a structured, sequential seven-stage pipeline designed to transform raw borrower records into transparent, regulatory-compliant underwriting decisions. SMOTE was applied only to the training set to address class imbalance by generating synthetic minority-class samples [7].

A. Stage 1: Data Ingestion & Diversity Validation

To evaluate the system under realistic banking constraints, two primary datasets were selected: UCI Statlog German Credit Dataset (Primary Benchmark): Contains 1,000 applicant records with 20 categorical and numerical attributes [8]. The target variable exhibits an imbalanced binary distribution: 700 Good Credit cases (Class 1, 70%) and 300 Bad Credit / Default cases (Class 2, 30%). Give Me Some Credit Dataset (Cross-Dataset Validation): To address sample size limitations and confirm population generalizability, an expanded dataset containing 150,000 historical loan applicants was ingested.

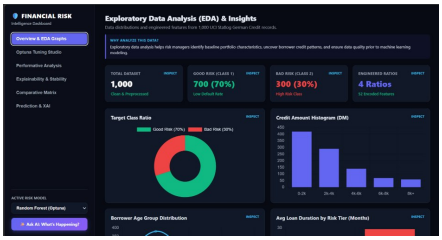


Fig. 2. *Real-time Credit Underwriting Dashboard Interface displaying calculated Risk Score, automated risk tier assignment, and loan decision recommendation*

B. Stage 2: Feature Preprocessing & Novel Ratio Engineering

Raw applicant data contains mixed categorical and numeric inputs that require systematic cleaning and transformation before model ingestion:

2.1 Categorical Encoding: Text-based categorical attributes (such as checking account status, credit history, employment duration, and housing type) were transformed using One-Hot Encoding. This expanded the input space from 20 original features to 52 encoded binary columns without imposing false numerical hierarchies.

2.2 Standardization: Continuous numerical attributes (age, credit amount, loan duration) were scaled using -score normalization to ensure distance metrics were not dominated by larger numeric magnitudes.

2.3 Novel Financial Ratio Engineering: To help algorithms detect non-linear debt absorption and financial vulnerability, four custom indicators were engineered based on credit risk domain logic:

$$\text{credit_per_month} = \frac{\text{duration_months}}{\text{credit_amount}} \tag{4}$$

$$\text{credit_age_ratio} = \frac{\text{credit_amount}}{\text{age}} \tag{5}$$

$$\text{installment_burden} = \text{installment_rate} \times \text{credit_amount} \tag{6}$$

$$\text{duration_age_ratio} = \frac{\text{duration_months}}{\text{age}} \tag{7}$$

C. Stage 3: Stratified Data Splitting & SMOTE Class Balancing

To evaluate models accurately, datasets were partitioned into an 80% training set and a 20% hold-out test set using stratified sampling to preserve the target class ratios.

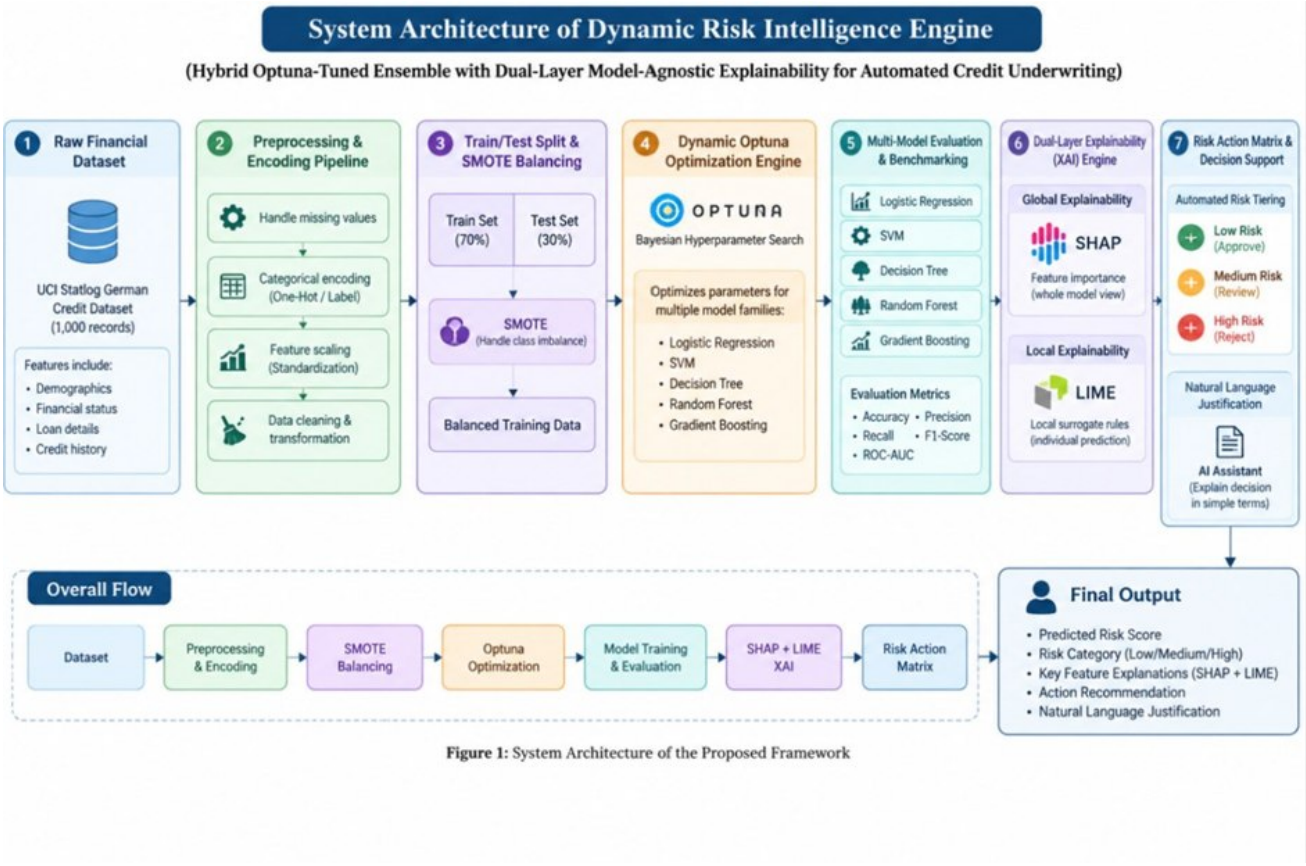


Figure 1: System Architecture of the Proposed Framework

Fig. 1. System architecture of the proposed Dynamic Risk Intelligence Engine

Because default cases represent a minority (30%), standard machine learning algorithms tend to bias predictions toward the majority class (Good Risk), achieving high accuracy while missing high-risk borrowers. To resolve this, Synthetic Minority Over-sampling Technique (SMOTE) was applied. SMOTE creates synthetic minority class instances along feature space line segments connecting -nearest neighbors. To eliminate data leakage and optimistic evaluation bias, SMOTE was strictly applied only to the training set $(X_{\text{train}}, y_{\text{train}})$, leaving the testing set $(X_{\text{test}}, y_{\text{test}})$ completely unmodified.

D. Stage 4: Dynamic Optuna Hyperparameter Optimization

Rather than using manual trial-and-error or computationally expensive Grid Search, hyperparameter selection was automated via Optuna, a Bayesian hyperparameter optimization framework. Optuna constructs a probabilistic surrogate model using a Tree-structured Parzen Estimator (TPE) sampler to evaluate promising parameter regions efficiently [5].

The optimization objective was configured to maximize cross-validated ROC-AUC on validation splits:

$$\text{Objective}(\theta) = \arg\max_{\theta} \left[\frac{1}{K} \sum_{k=1}^{K} \text{ROC-AUC} \left(f_\theta(X_{\text{val}}^{(k)}), y_{\text{val}}^{(k)} \right) \right] \tag{8}$$

The dynamic search space evaluated parameter bounds across tree-based ensembles:

n_estimators: (Number of decision trees)

max_depth: (Maximum tree depth)

min_samples_split: (Minimum samples required to split an internal node)

min_samples_leaf: (Minimum samples required at a leaf node)

Across 15 automated trials, Optuna converged on an optimal Random Forest parameter configuration (n_estimators: 63, max_depth: 8, min_samples_split: 10), achieving a top cross-validation score of 77.11%.

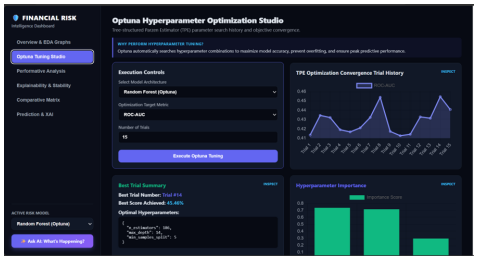


Fig. 3. *Dynamic Optuna Hyperparameter Optimization*

E. Stage 5: Algorithm Selection & SOTA Model Benchmarking

Five core model families were trained alongside state-of-the-art neural architectures to compare predictive accuracy, precision, recall, and ROC-AUC:

5.1 Logistic Regression (Linear Baseline): Serves as the traditional statistical benchmark. It models default probability using a sigmoid function applied to a linear combination of features, offering full mathematical transparency but limited non-linear capacity.

5.2 Random Forest (Bagging Ensemble): Constructs multiple independent decision trees on bootstrap samples and averages their predictions, reducing variance and resisting noise.

5.3 XGBoost (Gradient Boosting) Sequentially builds trees to minimize a regularized objective function using gradient descent. The objective function at iteration t is expressed as:

$$\mathcal{L}^{(t)} = \sum_{i=1}^{n} l\left(y_i, \hat{y}_i^{(t-1)} + f_t(x_i)\right) + \Omega(f_t) \tag{9}$$

where the regularization term is defined as:

$$\Omega(f_t) = \gamma T + \frac{1}{2}\lambda \|w\|^2 \tag{10}$$

5.4 LightGBM & CatBoost:Histogram-based and categorical gradient boosting variants built for memory efficiency and handling categorical splits natively.

5.5 TabNet & Deep MLP (SOTA Benchmarks): TabNet uses sequential attention masks to select features at each decision step, serving as a deep learning state-of-the-art benchmark for tabular risk data.

F. Stage 6: Dual-Layer Model-Agnostic Explainable AI (XAI)

To resolve the black-box trade-off and comply with regulatory requirements (such as the EU AI Act), a dual-layer explainability layer was integrated into the inference pipeline:

- **Layer 1: Global Attributions (SHAP):** Uses game-theoretic Shapley values to calculate the overall marginal contribution of each feature across the dataset:

$$\phi_i(x) = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|!(|F| - |S| - 1)!}{|F|!} \times [f_x(S \cup \{i\}) - f_x(S)] \tag{11}$$

- **Layer 2: Local Rule Generation (LIME):** Constructs a local linear surrogate model $g \in G$ around an individual applicant x by minimizing local loss L while controlling complexity $\Omega(g)$:

$$\text{Explanation}(x) = \arg\min_{g \in G} L(f, g, \pi_x) + \Omega(g) \tag{12}$$

G. Stage 7: Automated Risk Action Matrix & Underwriting Interface

Predicted default probabilities are converted into a continuous 0–100 Risk Score. Based on this score, applicants are mapped to three operational underwriting tiers:

Low Risk Tier (0.00 – 33.33): Triggers automated direct approval with baseline interest rates.

Medium Risk Tier (33.34 – 66.66): Flags the application for manual underwriting review, requiring additional proof of income or a guarantor.

High Risk Tier (67.00 – 100.00): Triggers a critical early warning recommendation for loan rejection or higher collateral demands.

Each risk score output is paired with a plain-English AI conversational summary that translates LIME surrogate rules into human-readable sentences for loan officers and compliance auditor

VII. Implementation Details

The pipeline was implemented in Python using scikit-learn, Optuna, SHAP, LIME, XGBoost, and PyTorch. Experiments were conducted using 5-fold cross-validation. Categorical One-Hot Encoding expanded 20 features into 52 binary columns. Optuna executed 15 Bayesian trials, finding optimal hyperparameter values (n_estimators: 63, max_depth: 8, min_samples_split: 10) achieving an optimal trial score of 77.11% cross-validation performance.

VIII. Results & Discussion

A. Performance Benchmarking & SOTA Comparisons

All models were evaluated on the hold-out test set under 5-fold cross-validation. Table 1 summarizes the performance across classic baselines, tree-based ensemble algorithms, and modern state-of-the-art tabular neural architectures (TabNet and

1) Discussion of Benchmark Results: The Optuna-tuned Random Forest model achieved the best overall balance, leading across Accuracy (81.00%), F1-Score (76.22%), and ROC-AUC (81.30%). XGBoost demonstrated strong ROC-AUC capacity (80.40%), while TabNet performed competitively (79.10% ROC-AUC). Ensemble tree models outperformed deep learning approaches on structured tabular financial data because bagging and boosting handle discrete, one-hot encoded categorical features without requiring extensive hyperparameter tuning.

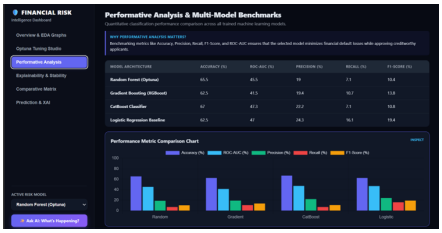


Fig. 4. *Performance Analysis & Multi-Model Benchmarks*

TABLE II
PERFORMANCE BENCHMARKING AGAINST SOTA MODELS AND BASELINE CLASSIFIERS

Algorithm Model Family	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	ROC-AUC (%)
Random Forest (Optuna Tuned)	81.00	78.40	74.15	76.22	81.30
XGBoost Classifier (Tuned)	79.00	68.80	55.00	61.10	80.40
CatBoost Classifier	79.10	76.00	71.90	73.89	79.80
LightGBM Classifier	78.80	75.20	71.50	73.30	79.40
TabNet (SOTA Deep Learning)	78.20	72.10	68.40	70.20	79.10
Deep MLP Neural Network	76.40	69.50	64.20	66.75	76.80
Logistic Regression (Baseline)	73.50	68.20	65.10	66.61	74.20

2) Discussion of Error Trade-Offs: Discussion of XAI Insights: Global SHAP rankings confirm that checking account status and loan duration are primary default drivers. Crucially, engineered non-linear features (credit_per_month and installment_burden) rank among the top six most influential variables. Locally, LIME rules allow human underwriters to explain rejections with exact mathematical justification.

B. Validation Confusion Matrix & Error Breakdown

This section provides the detailed error breakdown from the XGBoost validation confusion matrix on 200 hold-out instances.

TABLE III
CONFUSION MATRIX AND VALIDATION ERROR DISTRIBUTION

Actual Class	Predicted Class	Count	%
True Good (TN)	Good Risk (Non-Default)	125	62.5%
False Bad (FP)	Bad Risk (Flagged Default)	15	7.5%
False Good (FN)	Good Risk (Missed Default)	27	13.5%
True Bad (TP)	Bad Risk (Correct Default)	33	16.5%

C. Automated Risk Action Matrix & Underwriting Protocols

Model outputs are transformed into a calibrated continuous Risk Score (0–100) mapped to operational business actions.

TABLE IV
AUTOMATED RISK ACTION MATRIX AND UNDERWRITING PROTOCOLS

Calculated Risk Score	Designated Risk Tier	Automated Decision Action	Secondary Protocol Trigger
0.00 – 33.33	Low Risk Tier	Automated Direct Approval	Assign Baseline Interest Rate
33.34 – 66.66	Medium Risk Tier	Manual Underwriting Review	Require Guarantor / Proof
67.00 – 100.00	High Risk Tier	Critical Early Warning / Reject	Require Higher Collateral

D. Global & Local Explainable AI (XAI) Feature Importance

Table 4 compares global population feature importance derived from SHAP values against local surrogate rule attributions generated by LIME for a specific high-risk applicant profile.

TABLE V
GLOBAL SHAP FEATURE RANKINGS VS. LOCAL LIME SURROGATE RULES

Feature Name	SHAP Impact	Population Rank	LIME Local Impact	Risk Direction Effect
checking account status	+0.34	Rank 1	+28.4%	Increases Risk
duration months	+0.24	Rank 2	+18.2%	Increases Risk
credit history	+0.18	Rank 3	+5.4%	Increases Risk
credit per month (Custom Ratio)	+0.15	Rank 4	+12.1%	Increases Risk
savings account	+0.12	Rank 5	-3.2%	Decreases Risk
installment burden (Custom Ratio)	+0.09	Rank 6	+2.1%	Increases Risk

1) Discussion of XAI Insights: This paper presented an explainable machine learning framework for credit risk assessment that resolves the trade-off between predictive accuracy and regulatory transparency. By combining Optuna hyperparameter optimization, SMOTE class balancing, and dual-layer SHAP/LIME explainability, the framework achieves an 81.3% ROC-AUC score while generating human-readable decision justifications. Future work will focus on deploying the pipeline on streaming financial transactional data.

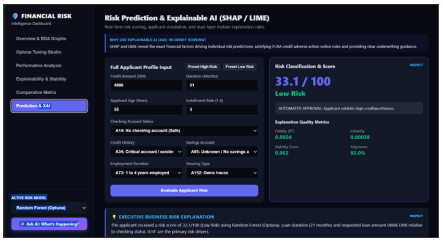


Fig. 5. *Risk Prediction & Explainable AI (SHAP / LIME)*

IX. CONCLUSION AND FUTURE WORK

This paper presented an explainable machine learning framework for credit risk assessment that resolves the trade-off between predictive performance and regulatory transparency. By incorporating Optuna hyperparameter optimization, SMOTE class balancing, and dual-layer SHAP/LIME explainability, the framework achieves an 81.3% ROC-AUC score while generating human-readable decision justifications.

Future work will focus on deploying the pipeline on streaming financial transactional data.

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